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Hydroxysafflor yellow A (HSYA) attenuates hypoxic pulmonary arterial remodelling and reverses right ventricular hypertrophy in rats

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ABSTRACT

Ethnopharmacological relevance: *Carthamus tinctorius* L. is a traditional herbal medicine native to China with properties of promoting blood circulation and removing blood stasis, which is used for the treatment of cerebrovascular and cardiovascular diseases. Hydroxysafflor yellow A (HSYA) is the main constituent isolated from the flower of *Carthamus tinctorius* L. which is used as a marker substance in the quality control of *Carthamus tinctorius* L. in Chinese Pharmacopeia.
Aim of the study: This study is to investigate the hypertension attenuating effect of HSYA on hypoxia-induced pulmonary artery hypertension model rats, and the possible mechanism.
Materials and methods: The animal models were made by treating adult male Wistar rats (of the same age with the same weight of 200 ± 25 g) under hypoxia 24 h per day for 9 days with or without administration of HSYA. The pulmonary arterial pressure of rats was measured after anesthetization; The right ventricular hypotrophy was evaluated by the right ventricular hypotrophy index (RVHI=[RV/(LV+S)]) as well as histomorphology assay with Hematoxylin and Eosin (HE) staining; The reducing of pulmonary artery remodelling was evaluated by histomorphology assay with HE staining; The proliferation of pulmonary artery smooth muscle cells (PASMCs) was evaluated by immunohistochemistry assays (PCNA and Ki67) and MTT assay. Cell cycle analysis and Weston-blot analysis were also performed in the study.
Results: HSYA reduced the mean right ventricular systolic pressure (RVSP) of rats with hypoxic pulmonary arterial hypertension (HPH) in a manner of concentration dependency. It significantly inhibited the PASMCs proliferation and attenuated the remodelling of the pulmonary artery and right ventricular hypertrophy.
Conclusion: These findings suggested that HSYA protected against hypoxic induced pulmonary hypertension by reversing the remodelling of the pulmonary artery through inhibiting the proliferation and hypertrophy of PASMCs. This is in accordance with our previous finding that HSYA protects against the pulmonary artery vascular constriction. All these results suggest that HSYA may be a promising candidate for HPH treatment.

1. Introduction

In recent years, clinical herbal medicine has drawn increasing attention for drug discovery. The flower of Carthamus tinctorius L. (English name Safflower) is a traditional Chinese herbal medicine (National Pharmacopoeia Committee, 2010) which has been extensively used for the treatment of cerebrovascular and cardio-vascular diseases for many years (Tien et al., 2010) and recorded in Chinese Pharmacopoeia. HSYA is a water soluble monomer isolated from safflower which is marked as a key active substance for the quality control of *Carthamus tinctorius* L. Research has shown that HSYA exerts a variety of effects upon the cardiovascular system. For example, HSYA can inhibit thrombosis and platelet aggregation (Tian et al., 2003), improve congestive cardiac failure in rats by suppressing ET-1 and iNOS and reduce oxidative stress in infarcted tissue (He et al., 2008). More recent studies have been reported that HSYA can present a therapeutic effect on Parkinson's disease (Han and Zhao, 2010),

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reduce phenylephrine or KCl-induced vasoconstriction (Zhang et al., 2011) and reduce blood pressure and heart rate (Nie et al., 2012). Our previous study (Bai et al., 2012) has shown that HSYA can protect against pulmonary arterial vasoconstriction in a way of adjusting the activity of K^+ channel in PASMCs. However, it remains unknown whether HSYA can lower the pulmonary hypertension in rats with PAH.

Pulmonary Arterial hypertension (PAH) is a devastating disease characterized by an abnormally sustained elevation of the blood pressure in the pulmonary circulation, leading to the right ventricle hypertrophy and dysfunction, with a high incidence of morbidity and mortality (Benisty et al., 2002). This is mainly caused by over -proliferation of the vascular wall cells, that will ultimately block small pulmonary arteries (Humbert et al., 2004; Pietra et al., 2004). At present, the definite mechanisms of this disease still remain unclear. A proper balance between proliferation and apoptosis is required for normal tissue homeostasis. When this balance is disturbed in favour of proliferation or too little apoptosis, the resultant increase in cell population can cause pathophysiological problems, as in pulmonary arterial hypertension (PAH) (Burg et al., 2008). HPH is the most common complication in cardiovascular diseases with severe increase of pulmonary arterial pressure, pulmonary vascular structure remodelling (PVSR) and right ventricular hypertrophy (RVH) resulting in heart failure and death without intervention (Naeije et al., 2014). In order to explore whether HSYA can be a treatment of PAH, in this paper, experiments were designed to test the influence of HSYA on the remodelling of the pulmonary artery vascular and the right ventricular hypertrophy in HPH rat models. Furthermore, to find out whether HSYA can improve the condition of HPH and what is the possible mechanism.

2. Material and methods

All the experiments were performed with the approval of the Harbin Medical University Ethic and Biosafety Committee (Harbin Medical University, Harbin, China). The study was performed in accordance with the Guide for the Care and Use of Laboratory Animals (National Research Institute, 1996). Mouse polyclonal antibody against proliferating cell nuclear antigen (PCNA), Rabbit polyclonal antibodies against α-smooth muscle actin (α-SMA) and β-actin were purchased from Beyotime Institute of Biotechnology, Shanghai, China; HSYA (purity > 98%, structure seen Fig. 1) was purchased from National Institute for Drug and Food Control, Harbin branch, specification: 20 mg/vial, batch No.: 20150149. Identification was done by ^{1}HNMR: ^{13}CNMR, all the spectrum data is in accordance with the reference (Feng et al., 2013).

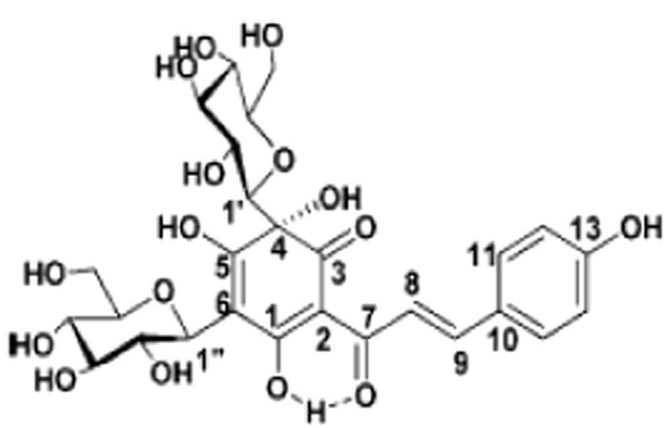

Fig. 1. Chemical structure of HSYA.

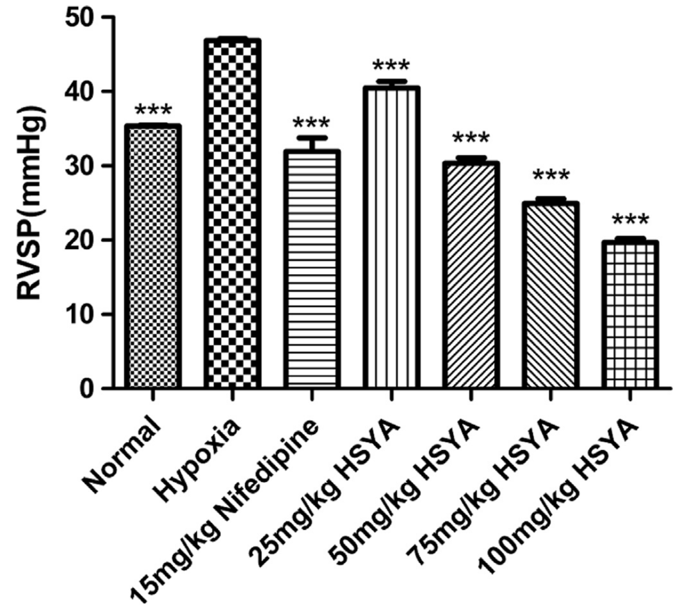

Fig. 2. Effect of HSYA on RVSP of rats with HPH *** indicates the result has statistical significance ($p < 0.005$, compared with Hypoxia).

Table 1
Effect of HSYA on RVH in rats with HPH.

	RV/ (LV+S)% RVHI
Control	23.20 ± 2.44
Hypoxia	37.87 ± 2.57*
Hypoxia+15 mg/kg Nifedipine	25.57 ± 2.58**
Hypoxia+25 mg/kg HSYA	33.55 ± 1.20*
Hypoxia+50 mg/kg HSYA	32.14 ± 2.14*
Hypoxia+75 mg/kg HSYA	29.42 ± 2.24*
Hypoxia+100 mg/kg HSYA	24.01 ± 1.18**

note: n =5. RV: right ventricle; LV: left ventricle; S: septum;
* $p < 0.05$ vs. control.
** $p < 0.01$ vs. Hypoxia.

2.1. Establishment of HPH rat models

Thirty five healthy male Wistar rats with the weight of 200 ± 25 g were supplied by Animal Experimental Centre of Harbin Medical University, Daqing Campus. Rats were divided into 7 groups (n=5 each) randomly and equally: (1) rats in hypoxia group were intragastrically administrated 2 mL/kg normal saline daily before being exposed to hypoxia; (2) Rats in control group were exposed to the normoxia and intragastrically administrated the same volume of normal saline at the same time each day as the model group; (3-6) Rats in HSYA treated group were intragastrically administrated HSYA 25 mg/kg, 50 mg/kg, 75 mg/kg, 100 mg/kg daily before being exposed to hypoxia, respectively; (7) Rats in positive control group were intragastrically administrated Nifedipine15 mg/kg. The rat models of HPH were established by exposing rats to a concentration of (10 ± 0.5) % O_2 24 hrs per day for 9 days.

2.2. Measurements of right ventricular systolic pressure

After 9 days of hypoxic treatment, rats were removed from hypoxia to recover for at least one hour, anesthetized with 2% sodium pentobarbital powder by injection (40 mg/kg) intraperitoneally, then followed pressure measurements. Animals were placed on a thermoregulated surgical plate. Right ventricular systolic pressure (RVSP) was measured with the use of a single lumen catheter (0.012" × 0.016" silicone tubing) through the right external jugular vein. Immediately after placement of the catheter into the right ventricle, RVSP was measured (Hu et al., 2002) with a Gould pressure transducer positioned at the mid-thorax of the animal and a Gould multichannel recorder. Proper catheter location was slightly verified by the waveform of the pressure tracing. After measurement of RVSP, all rats were sacrificed and used immediately for the determination of right ventricular hypertrophy.

2.3. Evaluation of right ventricular hypertrophy

After removal of atria, the hearts were dissected and then the right ventricle (RV) was separated from the left ventricle and septum (LV+S) under a dissecting microscope. The dissected wet samples were weighed after drying at 90 °C for 40 hr to obtain the ratio of RV/LV+S. The ratio of [RV/(LV+S)] was calculated as right

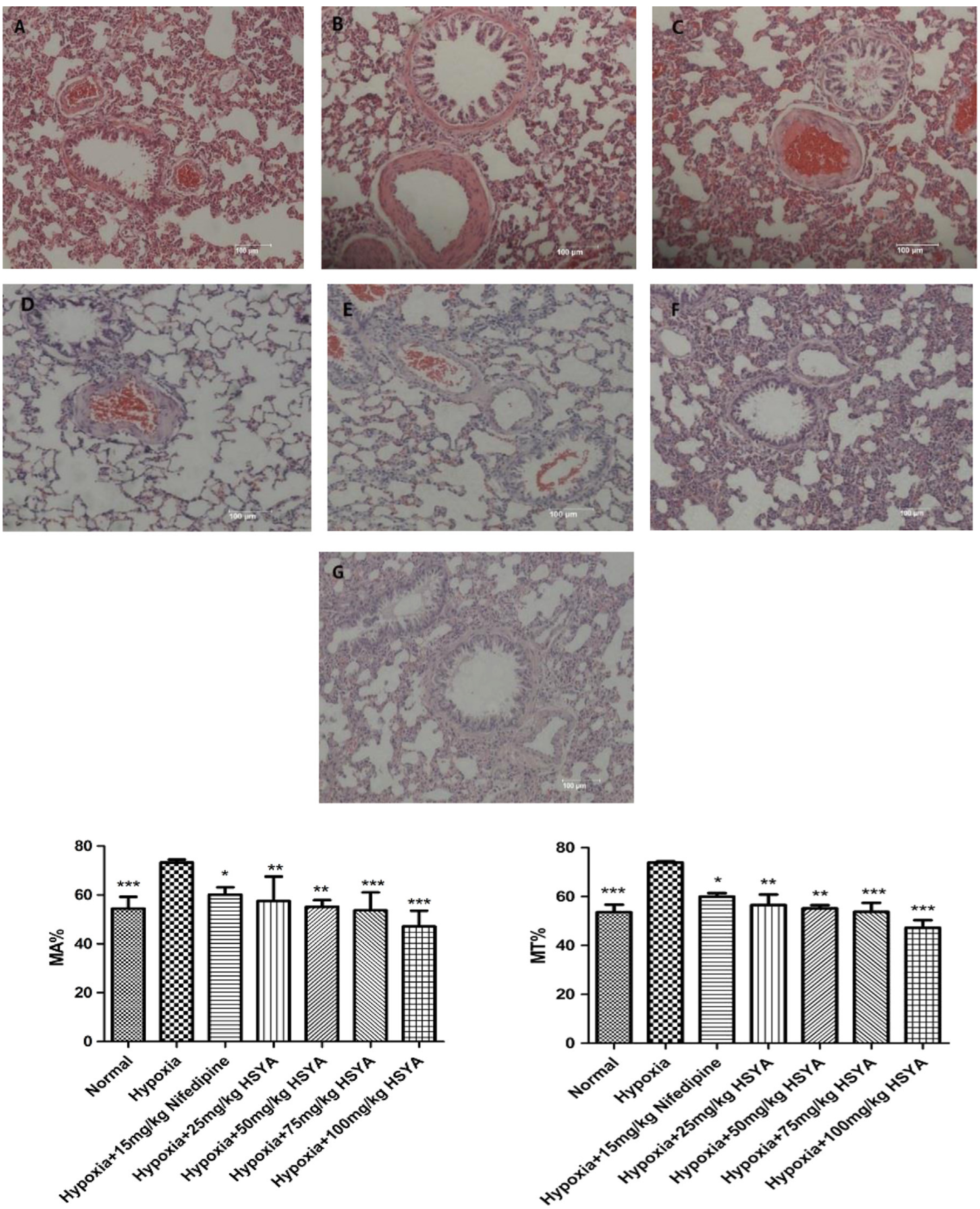

Fig. 3. Effects of HSYA on pulmonary artery morphology response in HPH rats (HE staining, at 200 × magnification). (A) normoxic group; (B) hypoxic group; (C) 25 mg/kg HSYA group; (D) 50 mg/kg HSYA group; (E) 75 mg/kg HSYA group; (F): 100 mg/kg HSYA group; (G): 15 mg/kg Nifedipine group. MT% means the ratio of vascular medial wall thickness to external diameter; MA% means the ratio of vascular medial cross-sectional area to total arterial cross-sectional area *indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.005$ vs. hypoxia.

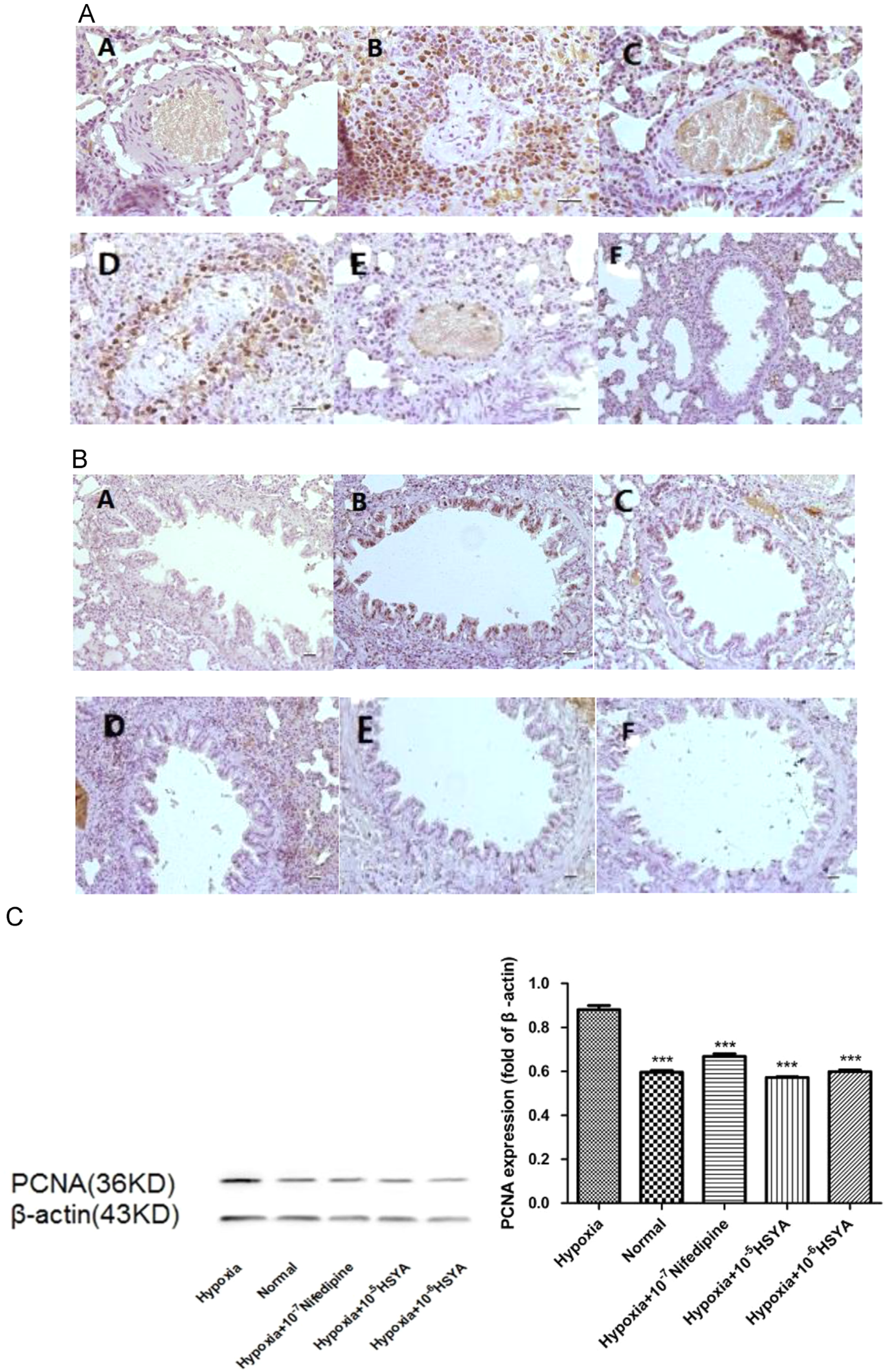

Fig. 4. A. Expression of PCNA-positive cells in HPH rat lung tissues (PCNA immunostaining, at 400 × magnification) (A): Normoxia group; (B): Hypoxia group; (C): Hypoxia + HSYA 25 mg/kg group; (D): Hypoxia + HSYA 50 mg/kg group; (E): Hypoxia + HSYA 75 mg/kg group; (F): Hypoxia + Nifedipine 15 mg/kg group. B. Expression of PCNA-positive cells in HPH rat lung small bronchioles (PCNA immunostaining, at 400 × magnification) A: Normoxia group; B: Hypoxia group; (C): Hypoxia + HSYA 25 mg/kg group; (D): Hypoxia + HSYA 50 mg/kg group; (E): Hypoxia + HSYA 75 mg/kg group; F: Hypoxia + Nifedipine group 15 mg/kg. C. The protein expression of PCNA in PASMCs cells with or without pretreatment with HSYA (HSYA (10^{-5} M, 10^{-6} M) under Hypoxia for 24 h. and Nifedipine (10^{-7} M) as a positive control. *** indicates $p < 0.005$ vs. hypoxic group.

ventricular hypertrophy index (RVHI) as described by Fulton (Fulton et al., 1952) to determine whether right ventricular hypertrophy occurred. The right lungs were immediately stored at −80 °C, whereas the left lungs were fixed in formalin for the following immunohistochemistry.

2.4. Histomorphometric analysis

Pulmonary artery morphology was performed as described previously (Dahal et al., 2010). Briefly, the lung tissues were embedded with paraffin and sliced serially after fixation in 10%

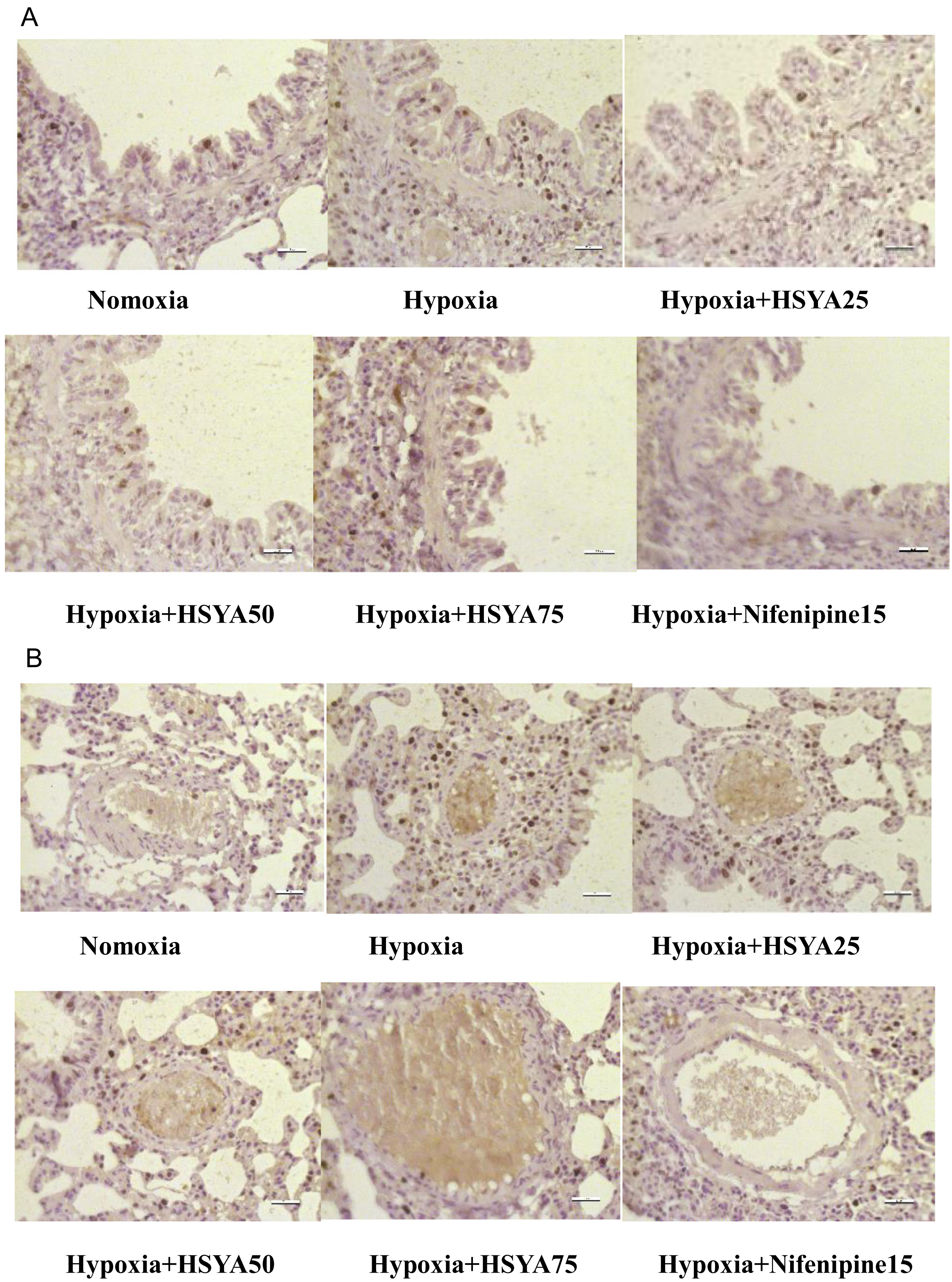

Fig. 5. A. Expression of Ki67-positive cells in Pulmonary Small bronchioles (Ki67 immunostaining, at 400× magnification). A: Normoxia group; B: Hypoxia group; (C): Hypoxia+HSYA 25 mg/kg group; (D): Hypoxia+HSYA 50 mg/kg group; (E): Hypoxia+HSYA 75 mg/kg group; F: Hypoxia+Nifedipine 15 mg/kg group. B. Expression of Ki67-positive cells in distal pulmonary arteries (Ki67 immunostaining, at 400× magnification). A: Normoxia group; B: Hypoxia group; (C): Hypoxia+HSYA 25 mg/kg group; (D): Hypoxia+HSYA 50 mg/kg group; (E): Hypoxia+HSYA 75 mg/kg group; F: Hypoxia+Nifedipine 15 mg/kg group.

formalin (pH 7.4), stained with HE. Morphologic images of the peripheral pulmonary artery and pulmonary small bronchioles were captured with a dissecting microscope and analysed using a computerized morphometric system. MT% (ratio of vascular medial wall thickness to external diameter) and MA% (ratio of vascular medial cross-sectional area to total arterial cross-sectional area) were calculated to assess the degree of PVSR.

2.5. Immunohistochemical analysis of Ki67 and PCNA

Paraffin-embedded lung tissue sections were de-paraffinized in xylene and rehydrated in a graded ethanol series to PBS (pH 7.4). Antigen retrieval was performed by heat at 100 °C in 0.01 M citrate buffer (pH 6.0) for 20 min. The sections were pre-treated with hydrogen peroxide (15%) to quench the activity of endogenous peroxidase following incubation with bovine serum albumin (10%) for 60 min and serum for 20 min in sequence; the sections were incubated with secondary antibody at 37 °C for 30 min after incubation with primary antibodies (rabbit polyclonal anti-PCNA and Ki67 respectively) at 4 °C overnight. Dyed with DAB and counterstained with hematoxylin, then cover slipped using mounting solution. PCNA-/Ki67-positive cells were counted throughout the entire section and the proliferation index (PI) was determined as the number of PCNA-/Ki67-positive cells per pulmonary vessel.

2.6. Isolation and culture of PASMCs

Wistar rats (200 ± 25 g) were decapitated and disinfected with 75% alcohol before the heart and lung tissues were removed. The lungs were rinsed with PBS buffer stored at 4 °C. The pulmonary arteries were split and cut into pieces (1–2 mm in length), placed onto the bottom of culture flasks, incubated and digested with collagenase type II for no more than 2 h in a 5% CO_2 cell culture incubator. Then DMEM (20% serum) was added. After 3–5 days, the cells were passaged till the cell density reached about 80–90%, and cells of passage 3–4 were subjected to serum starvation for 24 h before being used for the following experiments.

2.7. MTT assay for observing cell proliferation

The PASMCs were collected during the logarithmic growth phase and seeded in 96-well plates at a density of 5000 cells per well in 200 μL of culture medium. When reached 60–70% confluence, the cells were starved for 24 h to synchronize the cell cycles. After the culture medium being changed to 10% FBS, the PASMCs were cultured under normoxia and hypoxia (with or without HSYA) for 24 h respectively. At the end of the intervention, 20 μL of MTT solution (5 mg/mL) were added to each well and the plates were incubated for 4 h at 37 °C. Subsequently, 150 μL of DMSO was added to each well after the MTT solution was removed. The plates were agitated in darkness for 10 min. Immediately after the formed crystals fully dissolved, the absorbance value of each well was recorded at 490 nm wavelength by a microplate reader.

2.8. Flow cytometry assay

The PASMCs were trypsinized and centrifuged for 5 min at 3000 rpm. After being resuspended in 200 μL of PBS, the pellet was centrifuged and fixed in ice-cold ethanol (75%), stored at 4 °C overnight, and then centrifuged at the same speed and time as described above. After the final pellet being resuspended in 500 μL of PBS, RNase A was added, the so-formed mixture was incubated at 37 °C for 60 min and filtered through a flow tube. The PASMCs were then stained with propidium iodide at 4 °C in the dark for 30 min. Data was analysed by flow cytometry. The percentages of cells in S+G2/M phase was calculated to evaluate the changes in cell cycle distribution.

2.9. Western blotting analysis

PASMCs were washed 3 times in cold PBS, scraped into 0.3 mL of RIPA lysis buffer (0.2 mL) containing a complete protease inhibitor mixture and a protein phosphatase inhibitor, and incubated on ice for 30 min. Resulting cell lysates were sonicated and centrifuged at 13,500 rpm for 15 min at 4 °C. The supernatant was collected. After protein extraction, the PASMCs lysates were separated by sodium dodecyl sulphate polyacrylamide (SDSP) gel electrophoresis and transferred to Nitrocellulose Blotting Membranes. After 1 h of incubation in a blocking buffer (0.1% Tween 20% and 5% non-fat dry milk powder), the membranes were incubated with specific primary antibodies overnight at 4 °C. After incubation with the mouse polyclonal antibody against PCNA conjugated to horseradish peroxidase, super enhanced chemiluminescence detection reagents were used to detect the specific bands. The immunoblots were quantified by densitometric analysis using Gel image processing system. The monoclonal anti-β-actin antibody was used as a control.

2.10. Statistical treatment

All experimental data were analysed with SPSS software (version 17.0, SPSS Inc., USA). All data are presented as mean ± standard deviation (SD), and measurements from two groups were compared by independent t test. $P < 0.05$ was considered statistically significant.

3. Results

3.1. Hemodynamic during Hypoxia

HPH Rat models were successfully established by hypoxic methods.

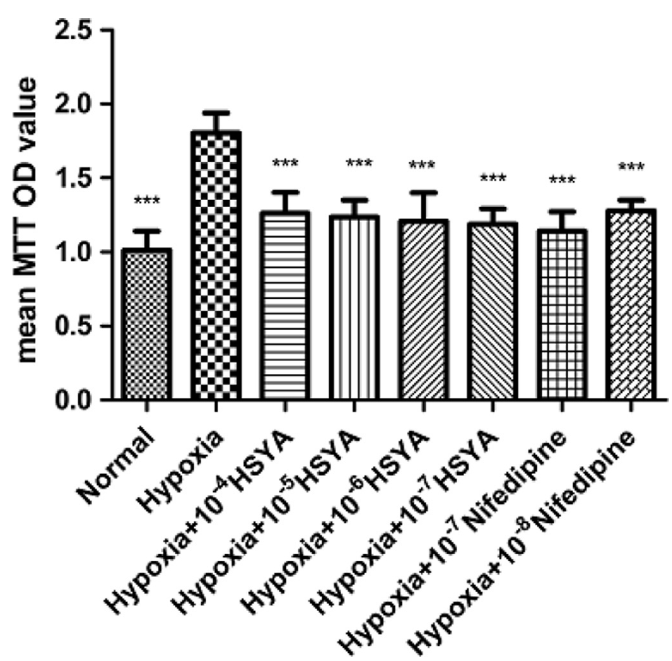

Fig. 6. Influence of HSYA on PASMCs viability under hypoxia. The cell viability was measured by MTT assay. PASMCs were incubated with or without HSYA (1×10^{-7}, 10^{-6}, 1×10^{-5}, 1×10^{-4} M) under hypoxia (10% O_2) or normoxia (21% O_2) for 24 h. All values are expressed as means ± S. D., Dunnet t-test was used to compare the differences between treated groups and control groups. *** $P < 0.005$ vs Hypoxia control.

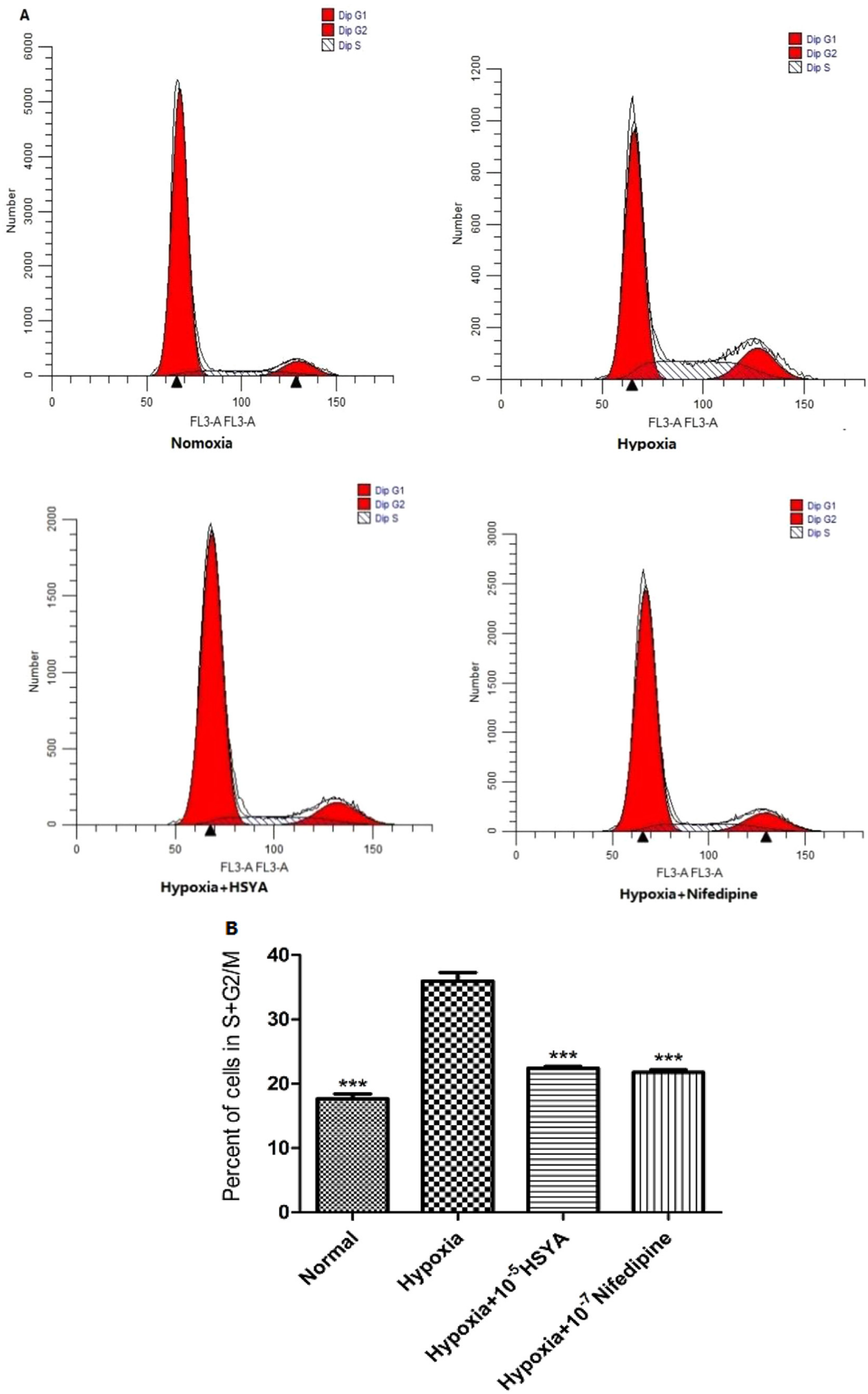

Fig. 7. Influence of HSYA on PASMCs cycle phase distribution under hypoxia. After serum starvation for 24 h, cells pre-treated or untreated with HSYA (1×10^{-5} M) were incubated under hypoxia or normoxia for 24 h. Cell cycle distribution was evaluated by flow cytometric analysis after nuclei staining with propidium iodide. Effect on G2/M and S phase was detected and statistically analysed. (A) Cell cycle distributions were recorded by flow cytometer in different treatments. A: Normoxia group; B: Hypoxia group; C: Hypoxia+HSYA 10^{-5} M group; D: Hypoxia+Nifedipine 10^{-7} M group. (B) The analysis of the proliferation index (PI) in rat PASMCs. The data are presented as means $\pm$ S. D. of four separate experiments. *** $P < 0.0005$ versus Hypoxia group.

Hemodynamic were measured as showed in Fig. 2. Compared with the normoxic group, acute hypoxia induced a significant increase in mean mRVSP (***$p < 0.005$ vs. control). Pre-treatment with the Ca^{2+} channel blocker Nifedipine markedly reduced the mRVSP during hypoxia (***$p < 0.005$ vs. hypoxia). Pre-administration of HSYA dose-dependently reduced mRVSP under hypoxia, respectively. The statistical difference was significant in the 50 mg/kg HSYA and 100 mg/kg HSYA groups (***$p < 0.005$ vs. hypoxia).

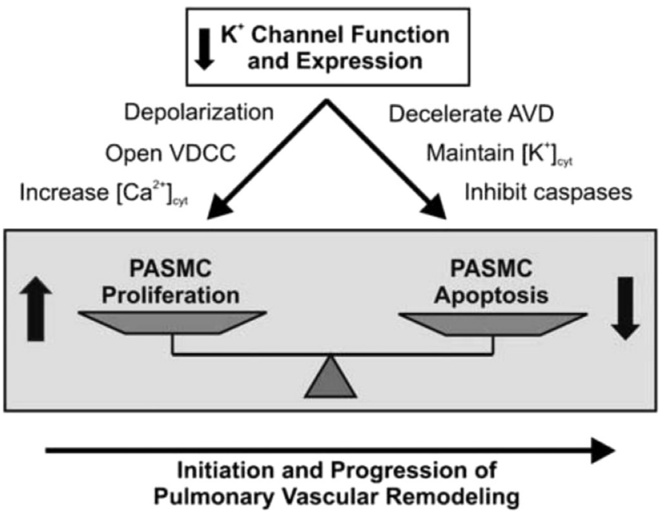

Fig. 8. Role of K+ channels in pulmonary vascular remodelling. Decreased K^+ channel function and expression not only stimulates pulmonary artery smooth muscle cell (PASMC) proliferation by increasing $[Ca^{2+}]$cyt, but also inhibits PASMC apoptosis by decelerating apoptotic volume decrease (AVD) and attenuating cytoplasmic caspase activity. The increased proliferation and inhibited apoptosisin PASMC may play an important role in initiation and/or progression of pulmonary vascular remodelling. VDCC, voltage dependent Ca^{2+} channels (Burg et al., 2008).

3.2. Effects on Pulmonary artery remodelling and RVH

Exposure to acute hypoxia induced pulmonary artery remodelling and right ventricular hypertrophy as reflected by increased MT%, MA% and RV/(LV+S)% in rats, respectively, shown in Table 1 and Fig. 3. The RVHI was increased significantly under hypoxia (*$p < 0.05$ vs. control); pre-administration with Nifedipine15 mg/kg, the RVHI was significantly decreased (**$p < 0.01$ vs. hypoxia) and pre-treatment with HSYA, the RVHI also reduced significantly and in a way of dose-dependency. (*$p < 0.05$, **$p < 0.01$ vs. hypoxia). In the same way, results in Fig. 3 showed that the MA% and MT% were decreased dose-dependently by pre-treated with HSYA (25 mg/kg, 50 mg/kg, 75 mg/kg and 100 mg/kg) whereas Nifedipine (15 mg/kg) was used as a positive control (*$p < 0.05$, **$p < 0.01$, ***$p < 0.005$ vs. hypoxia).

3.3. Effect of HSYA on inhibiting the proliferation of PASMCs

As shown in Fig. 4A and B, The PCNA -positive cells in HPH rat distal PAs and pulmonary small bronchioles significantly increased compared with normoxic group, and the percentage of PCNA-positive cells was significantly decreased compared with the hypoxia group by pre-treatment with HSYA in a dosage-dependent way (25 mg/kg, 50 mg/kg and 75 mg/kg), Nifedipine (15 mg/kg) as a positive control. And as we can see in Fig. 4C, the protein expression of PCNA in PASMCs was increased significantly under hypoxia (*** indicates $p < 0.005$ vs normal group); pre-treatment with HSYA (10^{-5} M, 10^{-6} M) significantly decreased the protein expression of PCNA in PASMCs (*** indicates $p < 0.005$ vs normal group), Nifedipine (10^{-7} M) as a positive control.

In Fig. 5A and B, the same influence of HSYA on HPH rats is shown both in pulmonary small bronchioles and distal Pas. The percentage of Ki67-positive cells was significantly decreased compared with the hypoxia group by pre-treatment with HSYA in a dosage-dependent way (25 mg/kg, 50 mg/kg and 75 mg/kg), Nifedipine (15 mg/kg) as a positive control.

3.4. Cell viability

As shown in Fig. 6, PASMCs viability increased significantly under hypoxia compared with normoxic group (*** $p < 0.005$), but the condition of high value of cell viability was significantly improved by pre-treatment with HSYA of different concentration (10^{-6}, 10^{-5}, 10^{-4} M), the effect is similar to the positive control (Nifedipine of 10^{-7}, 10^{-8} M) group. (*** $p < 0.005$ vs. hypoxic group).

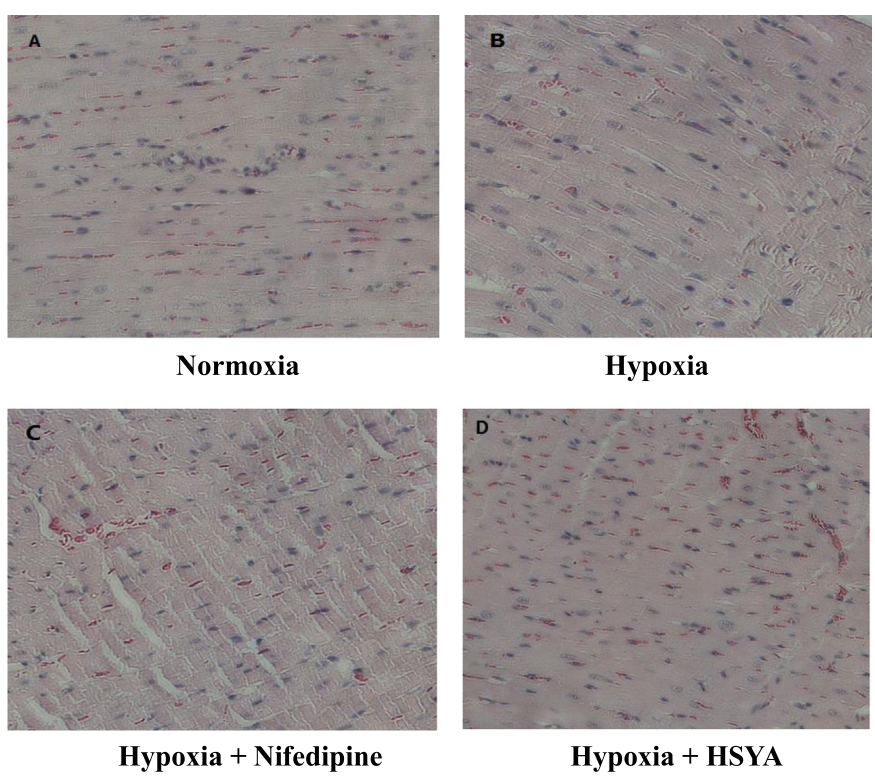

Fig. 9. The influence of HSYA on right ventricular hypotrophy in HPH rats (HE staining, at 200× magnification). (A) normoxic group; (B) hypoxic group; (C) Hypoxia+Nifedipine group; (D) Hypoxia+HSYA group.

3.5. Flow cytometry assay

As shown in Fig. 7, hypoxia caused PASMCs to accumulate in G2/M phase (from 17.68% to 35.96%) and in S phase (from 8.38% to 21.30%) compared with normoxia (***$p<0.005$). However, the number of cells in S phase significantly decreased from 21.30% to 9.92% by pre-treatment with 10^{-5} M of HSYA and from 21.30% to 10.97% by pre-treatment with 10^{-7} M of Nifedipine as a positive control, respectively. There was also a decreased percentage of cells in G2/M phases which was 22.37% and 22.06% respectively in HSYA group and Nifedipine group (***$p<0.005$ vs. hypoxic group).

4. Discussion

The main findings in this study are: 1) Pre-treatment with HSYA can significantly attenuate HPH. It significantly prevents the progression of pulmonary artery vascular remodelling, as well as the occurrence of right ventricular hypotrophy *ex vivo.* . 2) Treatment with HSYA significantly inhibits the proliferation of PASMCs in-vitro. This was evident from increased immunoreactivity of PCNA and Ki67 as well as the cell viability and the cell phase progression assays; 3) These findings are consistent with our previous report that showed HSYA can protect against pulmonary vascular constriction by activating the voltage-gated K^+ channel (Kv) in PASMCs in rats.

Extensive research to explore promising targets, and to develop effective therapeutic approaches for HPH has been undertaken over the past 20 years. Consequently, prolonged survival of patients and improvements of their lives has been achieved from the new therapeutic medications available for HPH. These medications include oxygen supplementation, diuretics, oral anticoagulation, and for a small percentage of patients calcium channel blockers. New treatments such as prostacyclin analogues, endothelin receptor antagonists, and phosphodiesterase type 5 inhibitors have been used in clinics (Chin et al., 2008; Boutet et al., 2008). Exploring new and better therapeutics, possible cures and pre-treatments are still drawing more and more attention. However, further advances are limited by the understanding of the pathogenic processes of HPH. This suggests that new hypotheses, observations, and studies might lead to new molecular targets for the treatment of HPH.

As shown in Fig. 8, Decreased K^+ channel function and expression not only stimulates PASMCs proliferation by increasing $[Ca^{2+}]$cyt, but also inhibits PASMC apoptosis. The increased proliferation and inhibited apoptosis in PASMC may play important roles in initiation and/or progression of pulmonary vascular remodelling. This suggests that K^+ channel as a therapeutic target for PAH remains an exciting possibility for future drug development (Burg et al., 2008).

The two main pathological processes of PAH are pulmonary vascular constriction and pulmonary vascular remodelling. The effect of HSYA against pulmonary vascular constriction has been previously proved (Bai et al., 2012). In this study, the main focus was on pulmonary vascular remodelling. Experiments including the histomorphology and immunochemistry assays were developed and the expecting results were obtained as indicated above.

To our knowledge, the pulmonary circulation is a high-flow, low-resistance and low-pressure system. Of which, the vascular resistance is determined by the contractility of the PASMCs of the medial layer of the pulmonary vascular wall. Sustained pulmonary vasoconstriction can cause increased pulmonary vascular resistance by narrowing the lumen of pulmonary arteries. Another major pathology of PAH is pulmonary vascular medial thickening. This is caused by PASMC hypertrophy and hyperplasia which results in the remodelling of the pulmonary vascular wall (Burg et al., 2008).

The criteria for the doses employed of the compound Hydroxysafflor yellow A was selected from reference to the Chinese Pharmacopeia 2010. This states that dependent on the patient's condition, the clinical dose of safflower is 15–20 mL/day (converted dosage of HSYA 75∼100 mg/day) by intravenous injection. Likewise in accordance to the formula of conversion between different species, the doses used for rats would be 9.1–12.2 mg/kg/day. However, our selected methodology for this study included intragastric administration. Therefore, the administration volume of HSYA was then assumed to be 50 mL/kg/day. Lower doses of 25 mg/kg/day and higher doses of 75 mg/kg/day and 100 mg/kg/day were also conducted in the study.

The results indicated that HSYA significantly inhibited pulmonary artery vascular remodelling while PASMCs hypertrophy and hyperplasia decreased. The PASMCs proliferation was inhibited significantly by HSYA as shown above. HSYA not only inhibited the remodelling of pulmonary artery, but also significantly inhibited the cell population in the small pulmonary bronchioles. This means that the complication in the small pulmonary bronchioles could be prevented by HSYA. We also found that the hypertrophy of cardiac muscle cells can also be prohibited by treatment of HSYA (shown in Fig. 9) and that this could be the reason why the right ventricular hypertrophy was attenuated by HSYA.

5. Conclusion

The results obtained in this study suggest that the marker constituent HSYA of *Carthamus tinctorius L.* possesses anti-hypertension activity in rats with HPH due to its reversing effect on pulmonary vascular remodelling via inhibiting the viability of PASMCs. HSYA also exhibits an anti-hypotrophy effect on the right ventricular of rats with HPH. It is possible that these effects can be attributed to the activation of the Kv channels in PASMCs. This effect was proven in our previous research. However, to interpret the precise molecular mechanism, further studies are required in future research. Together with the former research results, conclusion can be drawn that HSYA can be a potential candidate treatment for HPH by targeting the K^+ channel in PASMCs. This finding may also provide a partial interpretation of the traditional Chinese medicine *Carthamus tinctorius L.* that has therapeutic properties for promoting blood circulation and removing blood stasis.

Acknowledgments

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Glossary

HSYA: Hydroxysafflor yellow A;
PCNA: proliferating cell nuclear antigen;
PVSR: pulmonary artery structure remodelling;
PASMCs: pulmonary arterial smooth muscle cells;
HE staining: Harris Hematoxylin and Eosin Staining;
MT: medial wall thickness;
MA: medial cross-sectional area;
RV: right ventricular;
LV: left ventricle;
S: septum;
RVH: right ventricular hypertrophy;
DMEM: Dulbecco's modified Eagle medium;
PBS: phosphate buffer saline;
DAB, 3: 3-diaminobenzidine;
SDSP: sodium dodecyl sulphate polyacrylamide.